

PDR WORLD

OFFICIAL TECHNICAL SPECIFICATION MANUAL

ISO 9001:2015 • ISO 14001:2015 • RoHS • CACT

ENTERPRISE FIBER OPTIC & NETWORK INFRASTRUCTURE PLATFORM

Complete Master Knowledge Document & Technical Manual

Comprehensive architectural specification, complete 52-product catalog with detailed technical parameters, standardized 5-segment ordering code system, tactical drone optical solutions, and distributed acoustic/thermal sensing engineering guide.

50+

YEARS HERITAGE (1974)

3,000+

ENTERPRISE CLIENTS

52

PRODUCT FAMILIES

15+

EXPORT COUNTRIES

PDR Videotronics India Pvt. Ltd.

Corporate Office: 99 Old Prabhadevi Road, Mumbai –
400025, Maharashtra, India
Manufacturing Plant: Filmcity Complex, Goregaon East,
Mumbai – 400065

Website: <https://pdrworld.com> • **Email:**info@pdrworld.com / sales@pdrworld.com**Phone:** +91-22-24306494 • **WhatsApp:** +91 84199 16460

1. Executive Summary & Corporate Profile

PDR Videotronics India Pvt. Ltd. (branded globally as **PDR World**) is India's premier manufacturer and exporter of active and passive optical communication components, high-density fiber management systems, field test instruments, and specialty tactical optical solutions.

Corporate Philosophy: "Made in India. Deployed Worldwide."
PDR maintains 100% in-house manufacturing control across design, CNC precision polishing, cleanroom termination, and interferometric 3D testing.

Corporate Facilities Directory
Registered HQ: 99 Old Prabhadevi Rd, Mumbai 400025
Plant: Filmcity Complex, Goregaon East, Mumbai 400065
Sales & RFQs: sales@pdrworld.com • +91 84199 16460

50-Year Milestones
1974: Foundation in Mumbai for precision electronics.
1985: Established Fiber Optics Division.
2015: ISO 9001:2015 & exports to 15+ countries.
2025+: 400G Transceivers, Drone Optics & AI DAS/DTS.

2. Quality Assurance, Compliance & Certifications

CERTIFICATION	STANDARD & SCOPE	ACCREDITATION BODY
ISO 9001:2015	Quality Management System for Design, Manufacturing & Distribution	International Certification Services
ISO 14001:2015	Environmental Management System for Sustainable Clean Production	ICS Certified
RoHS Compliant	100% Free of Hazardous Substances (Lead, Mercury, Cadmium, Hex-Cr)	EU Directive 2011/65/EU
CACT Approval	Component Approval Centre for Telecommunications Approval	DoT (Department of Telecommunications)
IEC Standards	IEC 61300-3-35 (Endface cleanliness) & IEC 61754/61755 (Connectors)	International Electrotechnical Commission

100% Interferometric 3D Geometry & IL/RL Factory Inspection
Every connector undergoes 3D interferometry verifying Radius of Curvature (ROC), Apex Offset (<50µm), and Fiber Undercut/Protrusion per Telcordia GR-326-CORE, plus dual-wavelength (1310/1550nm) Insertion & Return Loss testing.

3. Critical Infrastructure & Smart Sensing Systems

PDR engineers mission-critical optical sensing, monitoring, and telemetry platforms designed for defense, oil & gas, utilities, and high-security critical infrastructure:

Remote Fiber Monitoring (RFMS)

Automated, intelligent continuous fiber plant surveillance with zero human intervention.

- **Switching Speed:** < 10ms optical path failover rerouting.
- **GIS Pinpointing:** Real-time fault location to ±1m on dark and live fibers.
- **Protocols:** Direct SCADA, NMS, and SNMP NOC telemetry alarms.

Distributed Acoustic Sensing (DAS)

Turns standard single-mode optical fibers into high-sensitivity distributed microphones.

- **Range:** 50 km+ per channel with ±5m spatial resolution.
- **AI/ML Detection:** Classifies footsteps, vehicles, manual trenching & pipeline leaks.
- **Infrastructure:** Border security, pipeline monitoring, and high-speed rail tracks.

Distributed Temperature Sensing (DTS)

Raman-scattering based continuous thermal profiling along linear infrastructure.

- **Precision:** ±1°C thermal resolution along 10km+ links.
- **Scan Speed:** < 10s fast multi-zone acquisition cycle.
- **Protection:** High-voltage power cables, conveyor belts & tunnel fire alarms.

Tactical Drone Optical Tether Kit

Ultra-lightweight optical link providing jam-proof high-bandwidth tethering for UAVs.

- **Jam-Proof:** 100% immune to RF EW guns, jamming, and electronic spoofing.
- **Hardware:** 5–20km micro-fiber spool (<1.2kg), <35g Sky Unit.
- **Latency:** <1ms uncompressed 4K low-latency video to ground tactical base.

Integrated Sensing Architecture

All PDR sensing interrogators interface with enterprise GIS maps, SCADA consoles, and automated optical bypass switches to maintain zero-downtime uptime for telecom backbones and tactical defense operations.

4. Complete 52-Product Technical Catalog

Comprehensive technical specifications, features, and parameters for all 52 products categorized into 6 divisions:

4.1 Active Components & Optical Transceivers (14 Products)

#1 Active Optical Cable (AOC)

DYNAMIC SPECS

Active Components > Active Optical Cable (AOC) • Slug: aoc

Explore high-performance Active Optical Cable (AOC) engineered by PDR World for enterprise telecom and data center applications.

KEY FEATURES:

- Hot-pluggable electrical interface with low power consumption
- Available in 10G SFP+, 25G SFP28, 40G QSFP+, 100G QSFP28, and 400G QSFP-DD
- Pre-terminated OM3/OM4 multimode fiber for high-speed inter-rack connectivity
- Lighter weight and smaller bend radius than copper DAC cables

SPECIFICATIONS:

Data Rate	10G / 25G / 40G / 100G / 400G
Form Factor	SFP+, SFP28, QSFP+, QSFP28, QSFP-DD
Cable Type	Multimode Fiber (OM3 / OM4)
Operating Temperature	0°C to 70°C (Commercial)
Power Consumption	< 1.5W per transceiver end
Standard Compliance	MSA & IEEE 802.3 standards

#2 Direct Attached Cable (DAC)

DATASHEET AVAILABLE

Active Components > Direct Attached Cable (DAC) • Slug: dac

PDR SFP+ Direct Attach Cables are passive, hot-swappable, MSA-compliant 10 Gbps twinax copper cables for reliable short-reach connectivity, factory-tested and RoHS/REACH compliant.

KEY FEATURES:

- Passive twinax copper, zero power, no heat generation
- Available in 1 m, 2 m, 3 m and 5 m lengths
- Supports 10 Gbps (10GBASE-CR, 10G FCoE, 10G InfiniBand)
- Hot-swappable SFP+ connectors, SFP+ MSA compliant

SPECIFICATIONS:

Cable Type	Passive twinax copper
Connector Type	SFP+ (both ends)
Nominal Data Rate	10.3125 Gbps
Available Lengths	1 m 2 m 3 m 5 m
Link Distance	Up to 7 m (passive)
Minimum Bend Radius	30 mm
Operating Temperature	0°C to 70°C
Power Consumption	Passive, zero power draw
Compliance	SFP+ MSA, IEEE 802.3ae, 10GFC, RoHS/REACH, TÜV/UL

#3 Bypass Switch

Active Components > Bypass Switch • Slug: bypass-switch

DATASHEET AVAILABLE

Carrier-grade 1U optical path protection that reroutes traffic around a failed inline device in under 50 ms via power, optical-power, and TCP/IP heartbeat monitoring, with up to 4 channels and dual AC/DC power.

KEY FEATURES:	SPECIFICATIONS:																
- Automatic bypass protection via power supply, optical power, and TCP/IP heartbeat monitoring — individually or in combination - Sub-50 ms carrier-grade switching speed for transparent network failover - Up to 4 independent bypass channels in a single 1U 19-inch rack chassis - Auto-Switch and Manual-Switch modes with configurable auto-return after an idle period	<table><tr><td>Working Wavelength</td><td>1310 nm / 1550 nm (selectable)</td></tr><tr><td>Fiber Type</td><td>SM 9/125 μm (ITU-T G.652)</td></tr><tr><td>Channels</td><td>Up to 4 independent channels</td></tr><tr><td>Insertion Loss</td><td>≤ 1.5 dB</td></tr><tr><td>Return Loss</td><td>≥ 50 dB</td></tr><tr><td>Switching Time</td><td>< 50 ms</td></tr><tr><td>Power Supply</td><td>AC 85–264 V DC 36–72 V (dual inputs)</td></tr><tr><td>Dimensions</td><td>19-inch rack, 1U height</td></tr></table>	Working Wavelength	1310 nm / 1550 nm (selectable)	Fiber Type	SM 9/125 μm (ITU-T G.652)	Channels	Up to 4 independent channels	Insertion Loss	≤ 1.5 dB	Return Loss	≥ 50 dB	Switching Time	< 50 ms	Power Supply	AC 85–264 V DC 36–72 V (dual inputs)	Dimensions	19-inch rack, 1U height
Working Wavelength	1310 nm / 1550 nm (selectable)																
Fiber Type	SM 9/125 μm (ITU-T G.652)																
Channels	Up to 4 independent channels																
Insertion Loss	≤ 1.5 dB																
Return Loss	≥ 50 dB																
Switching Time	< 50 ms																
Power Supply	AC 85–264 V DC 36–72 V (dual inputs)																
Dimensions	19-inch rack, 1U height																

#4 Optical Line Protection System (OLPS)

Active Components > Optical Line Protection System (OLPS) • Slug: olps

DATASHEET AVAILABLE

PDR Optical Line Protection System (OLP): Compact 1U optical-layer protection with real-time monitoring, automatic fiber switching, 1+1/1:1/1-1 protection, redundant AC/DC power, and NMS support.

KEY FEATURES:	SPECIFICATIONS:																
- Compact 1U, 19-inch rack-mountable form factor - Automatic fiber line switching on failure or degradation - Real-time optical power monitoring of both channels - Three protection architectures: 1+1, 1:1 and 1-1	<table><tr><td>Protection Architecture</td><td>1+1 1:1 1-1</td></tr><tr><td>Form Factor</td><td>1U rack-mountable (19-inch)</td></tr><tr><td>Working Wavelength</td><td>1310 nm or 1550 nm (selectable)</td></tr><tr><td>Switching Threshold</td><td>Default -30 dBm (configurable)</td></tr><tr><td>Switching Delay</td><td>Immediate or 1 to 999 s</td></tr><tr><td>Auto Switchback</td><td>Configurable 1 to 999 minutes</td></tr><tr><td>Management</td><td>RJ-45 Ethernet RS-232 DB9</td></tr><tr><td>Power Supply</td><td>220V AC -48V DC (dual redundant)</td></tr></table>	Protection Architecture	1+1 1:1 1-1	Form Factor	1U rack-mountable (19-inch)	Working Wavelength	1310 nm or 1550 nm (selectable)	Switching Threshold	Default -30 dBm (configurable)	Switching Delay	Immediate or 1 to 999 s	Auto Switchback	Configurable 1 to 999 minutes	Management	RJ-45 Ethernet RS-232 DB9	Power Supply	220V AC -48V DC (dual redundant)
Protection Architecture	1+1 1:1 1-1																
Form Factor	1U rack-mountable (19-inch)																
Working Wavelength	1310 nm or 1550 nm (selectable)																
Switching Threshold	Default -30 dBm (configurable)																
Switching Delay	Immediate or 1 to 999 s																
Auto Switchback	Configurable 1 to 999 minutes																
Management	RJ-45 Ethernet RS-232 DB9																
Power Supply	220V AC -48V DC (dual redundant)																

#5 Smart SFP Transceiver

Active Components > SFP Transceivers • Slug: smart-sfp

DATASHEET AVAILABLE

Integrated OAM (Operations, Administration, and Maintenance)

KEY FEATURES:	SPECIFICATIONS:								
- Integrated OAM (Operations, Administration, and Maintenance) - Real-time performance monitoring - Micro-loopback capability	<table><tr><td>Interface</td><td>1G / 10G SFP</td></tr><tr><td>Protocol</td><td>IEEE 802.3ah / 802.1ag</td></tr><tr><td>Diagnostics</td><td>DDM/DOM</td></tr><tr><td>Latency</td><td>Ultra-Low</td></tr></table>	Interface	1G / 10G SFP	Protocol	IEEE 802.3ah / 802.1ag	Diagnostics	DDM/DOM	Latency	Ultra-Low
Interface	1G / 10G SFP								
Protocol	IEEE 802.3ah / 802.1ag								
Diagnostics	DDM/DOM								
Latency	Ultra-Low								

#6 400G**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-400g](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#7 100G & 100G BIDI**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-100g-bidi](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#8 40G**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-40g](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#9 25G & 25G BIDI**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-25g-bidi](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#10 SFP+ 10G BIDI**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-10g-bidi](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#11 SFP+ 10G Dual Fiber**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-10g-dual](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#12 1G SFP BIDI**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-1g-bidi](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#13 1G SFP Dual Fiber**DATASHEET AVAILABLE****Active Components > SFP Transceivers • Slug: [sfp-1g-dual](#)**

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

#14 Copper SFP+

DATASHEET AVAILABLE

Active Components > SFP Transceivers • Slug: [sfp-copper](#)

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

KEY FEATURES:

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles

SPECIFICATIONS:

SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

4.2 Passive Components, Assemblies & WDM Systems (18 Products)**#15 Fiber Optic Patch Cords & Pigtailed**

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: [fo-patchcords](#)

PDR Fiber Optic Patch Cords and Pigtailed offer IEC/EIA/TIA/GR-326 compliant, factory-terminated singlemode and multimode assemblies with multiple connector, cable, and fiber options.

KEY FEATURES:

- Widest connector selection: FC, DIN, LC, MU, MTRJ, SC, ST, MPO, D4, Biconic, SMA, E-2000
- Singlemode fibers: OS2, G.657A1, G.657A2, multimode fibers: OM1, OM2, OM3, OM4
- Simplex, duplex, and fanout (up to 24F) configurations; 0.9 mm, 2.0 mm, and 3.0 mm cable diameters
- Low insertion loss ≤ 0.3 dB (PC) for both SM and MM; high return loss ≥ 50 dB UPC, ≥ 60 dB APC

SPECIFICATIONS:

Cable Diameter	0.9 mm 2.0 mm 3.0 mm (or custom)
Cable Configuration	Simplex Duplex Fanout (up to 24F)
SM Fiber Type	OS2 G.657A1 G.657A2
MM Fiber Type	OM1 OM2 OM3 OM4
Connector Types	FC LC SC ST MPO MU MTRJ DIN D4 Biconic SMA E-2000
Polish	UPC APC PC
SM Insertion Loss (PC)	≤ 0.3 dB
SM Return Loss, UPC	≥ 50 dB
SM Return Loss, APC	≥ 60 dB
MM Insertion Loss (PC)	≤ 0.3 dB
MM Return Loss (PC)	≥ 35 dB
Exchangeability	≤ 0.2 dB
Repeatability	≤ 0.2 dB
Operating Temperature	-20°C to +75°C

#16 MPO Cable Assembly

Passive Components > Cable Assembly • Slug: mpo-assembly

DATASHEET AVAILABLE

PDR MPO Cable Assemblies are high-density 4–24 fiber OS2/OM3/OM4/OM5 trunk cables with MT ferrules, ribbon or round cables, and male/female PC/APC connectors for fast data center deployment.

KEY FEATURES:

- High-density MT ferrule with precision guide pins for reliable multi-fiber alignment in 4, 8, 12, and 24F configurations
- Fiber options: OS2 (singlemode), OM3, OM4, OM5 (multimode), supports 10GbE, 40GbE, and 100GbE migration
- Cable types: jacketed ribbon, bare ribbon, and round cable in single and double jacket constructions
- End-face polish: PC and APC; connector housings available in Green, Yellow, Beige, and Aqua colours

SPECIFICATIONS:

Fiber Mode	SM: OS2 MM: OM3, OM4, OM5
Fiber Count	4F 8F 12F 24F 48F 72F 96F 144F Custom
Cable Type	Jacketed Ribbon Bare Ribbon Round
Cable Material	LSZH PVC
Cable Diameter, Single Jacket 12F	3.0 mm
Cable Diameter, Double Jacket 12F/24F	4.5 mm
Tensile Strength, Single 12F (short/long)	200 N / 100 N
Tensile Strength, Double 12F/24F (short/long)	400 N / 150 N
Crush Resistance, Single 12F	500 N/100 mm
Crush Resistance, Double 12F	1,000 N/100 mm
Min. Bending Radius	10×D (installed) 20×D (under load)
Cable Operating Temperature	, 20°C to +70°C
MPO Connector Type	Male or Female; PC or APC end face
Available Ferrule Counts	4, 8, 12, 24

#17 FanOut Patch Cords

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: fanout-patch-cords

PDR MTP/MPO Fanout Patch Cord enables high-density MPO/MTP-to-single-fiber breakout for up to 100G data center applications, available in multiple cable, connector, and fanout configurations.

KEY FEATURES:

- Small form factor MPO/MTP breakout to LC, SC, ST, or FC pigtails in 0.9 mm, 2.0 mm simplex, duplex, or Uniboot formats
- Available in 8F, 12F, 24F, 48F, 72F, 96F, and 144F fiber counts; fully custom solutions on request
- Supports single-mode (OS1, OS2) and multimode (OM1, OM2, OM3, OM4) fiber types
- MTP/MPO connectors available in Standard and Elite (low-loss) versions for flexible power budget management

SPECIFICATIONS:

MPO Insertion Loss, Typical	SM Standard: 0.35 dB SM Low Loss: 0.20 dB MM Standard: 0.35 dB MM Low Loss: 0.20 dB
MPO Insertion Loss, Maximum	SM Standard: 0.75 dB SM Low Loss: 0.35 dB MM Standard: 0.60 dB MM Low Loss: 0.35 dB
LC / SC / ST / FC Insertion Loss, Typical	SM Standard: 0.3 dB SM Low Loss: 0.2 dB MM Standard: 0.3 dB MM Low Loss: 0.2 dB
Return Loss	≥60 dB (APC) ≥20 dB (PC) ≥50 dB (UPC)
Durability	< 0.2 dB change after 200 insertions
Operating Temperature	–40°C to +75°C
Fiber Mode, Single-Mode	OS1, OS2
Fiber Mode, Multimode	OM1, OM2, OM3, OM4
MPO Connector Type	Male or Female; PC or APC end face
Single Ferrule Connector Types	LC, SC, ST, FC; PC or APC end face
Available Ferrule Counts	4F, 8F, 12F, 24F
Fiber Count Options	8F, 12F, 24F, 48F, 72F, 96F, 144F; custom available
Cable Jacket Diameter	0.9 mm 2.0 mm 3.0 mm 4.5 mm 7.0 mm 8.0 mm
Cable Construction	Micro single jacket Micro double jacket

#18 FTTX Smart Bullet Drop Cable

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: rapid-push

Single-fiber SC/APC drop assembly with a Ø5.9 mm bullet head for traction pulling through narrow ducts, ≤0.3 dB IL, ≥60 dB RL over 1260–1650 nm, 100% optically tested.

KEY FEATURES:

- Bullet-shaped connector head (Ø5.9 mm) with guide-wire hole for traction installation through narrow ducts
- Pull-through design for free and occupied ducts, no large bore holes, conduits, or specialist tools required
- SC/APC connectors with zirconia ceramic ferrule: insertion loss ≤0.3 dB (typ. 0.2 dB), return loss ≥60 dB
- Wide operating wavelength 1260 to 1650 nm covers all standard telecom bands including O, E, S, C, L, and U

SPECIFICATIONS:

Connector Type	SC/APC bullet-shaped (near-end) / SC/APC (far-end)
Bullet Connector Diameter	Ø5.9 mm
Operating Wavelength	1260 to 1650 nm
Insertion Loss	≤0.3 dB (typ. 0.2 dB)
Return Loss	≥60 dB
Tractive Force	≥150 N
Working Temperature	–40°C to +70°C

#19 CPRI Patchcord

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: cpri-patchcord

PDR CPRI Outdoor Waterproof Patch Cords feature rugged IP-rated, blind-mate fiber assemblies with multiple waterproof connector options, armored/non-armored cable designs, and 2–24 core single-mode or multimode fiber for harsh outdoor applications.

KEY FEATURES:

- Waterproof connector housings: ODVA, IP, FX, PDLC, and NSN types with MPO/MTP, LC, or SC ferrules
- IP-rated blind-mating design for quick, tool-free installation in field and outdoor environments
- Robust bayonet locking mechanism for easy, fast, and secure connector mating and retention
- Lightweight, waterproof, dustproof, and corrosion-resistant construction for harsh environments

SPECIFICATIONS:

Cable Diameter	5.0 mm 7.0 mm 8.0 mm (±0.3 mm)
Connector Housing Types	ODVA IP FX PDLC NSN
Ferrule / Connector Types	MPO/MTP LC SC
Fiber Mode, Single-Mode	G.652D G.657A1 G.657A2 G.657B3
Fiber Mode, Multimode	OM1 OM2 OM3 OM4
Fiber Count	2 / 4 / 8 / 12 / 24 core
Cable Jacket	LSZH TPU PE
Cable Length	1 to 150 m; custom lengths available
Operating Temperature	–40°C to +85°C
SM Insertion Loss, Typical	≤0.30 dB (IEC 61300-3-34)
SM Insertion Loss, Maximum	≤0.75 dB (IEC 61300-3-34)
MM Insertion Loss, Typical	≤0.50 dB (IEC 61300-3-34)
MM Insertion Loss, Maximum	≤0.75 dB (IEC 61300-3-34)
SM Return Loss	≥50 dB (PC) ≥60 dB (APC) (IEC 61300-3-6)

#20 Mode Conditioning Patchcord

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: mode-conditioning

Offset-launch patchcord that runs single-mode LX/LH laser sources over legacy 62.5/125 or 50/125 multimode fiber, eliminating Differential Mode Delay for reliable Gigabit Ethernet.

KEY FEATURES:

- Ensures Gigabit transmission over multimode fiber
- Eliminates Differential Mode Delay (DMD)
- Offset launch technique from single-mode to multimode
- Compatible with 62.5/125 μm or 50/125 μm MMF cable plant

SPECIFICATIONS:

Cable Plant	62.5/125 or 50/125 Multimode Fiber (MMF)
Insertion Loss (MMF–SMF)	≤ 2.1 dB (50/125); ≤ 2.6 dB (62.5/125)
Return Loss	≥ 50 dB (UPC); ≥ 65 dB (APC)
Durability	≤ 0.20 dB typical after 500 matings
Operating Temperature	–20 °C to +75 °C

#21 SMPTE Cable Assembly

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: smpte-assembly

Ruggedized, dirt-protected SM/MM fiber assemblies with all-metal housings, heavy-duty retention, and silicon gaskets, engineered for HDTV broadcast and outdoor OB applications

KEY FEATURES:

- Rugged all-metal connector housing for durability
- Heavy-duty cable retention for secure long-term performance
- Silicon gasket protection against dust and dirt
- Wide connector range: SC, LC, FC, ST, MU, MTRJ, SMA, E2000

SPECIFICATIONS:

Fiber Type	Single-mode (9/125 μm); Multimode (50/125, 62.5/125, OM3)
Connector Options	SC, LC, FC, ST, MU, MTRJ, SMA, E2000
Insertion Loss	≤ 0.30 dB typical; ≤ 0.50 dB max
Return Loss	≥ 50 dB UPC; ≥ 60 dB APC
Length Options	0.5 m to 999 m (custom available)
Operating Temperature	–40 °C to +75 °C

#22 POF — Plastic Optical Fiber Patchcord

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: `pof-patchcord`

PDR Plastic Optical Fiber (POF) Cable Assemblies provide cost-effective, EMI-immune connectivity with SI-POF cables, LC/RedLink™/VersaLink® patch cords, reel options for custom installation, UL VW-1 jackets, and compatibility with standard POF transceivers.

KEY FEATURES:

- 1 mm step-index SI-POF with numerical aperture 0.5, compliant with IEC 60793-2-40 Class A4a
- Low attenuation: 0.19 dB/m at 650 nm, ideal for short-range links up to tens of metres
- Simplex cable OD 2.2 mm; duplex (figure-of-8) minor axis 2.2 mm, major axis 4.4 mm
- UL VW-1 flame-retardant jacket option available for building and equipment installations

SPECIFICATIONS:

Fiber Standard	IEC 60793-2-40 Class A4a (SI-POF)
Numerical Aperture	0.5
Attenuation at 650 nm	≤0.19 dB/m
Fiber Diameter (incl. cladding)	0.94 to 1.06 mm (nominal 1.0 mm)
Cable Diameter, Simplex	2.13 to 2.27 mm (nominal 2.2 mm)
Cable Diameter, Duplex Minor Axis	2.13 to 2.27 mm
Cable Diameter, Duplex Major Axis	4.3 to 4.5 mm
Jacket Material	Flame-resistant PE or PVC
Jacket Color	Black; Duplex cable has white identifying bar on one strand
UL Flame Rating	VW-1
Tensile Stress, Simplex / Duplex	70 N / 140 N
Minimum Bend Radius	25 mm
Repeat Bending Endurance	≥10,000 cycles (≤1 dB additional loss)
Operating Temperature	-40°C to +85°C

#23 CAT 6 Patch Cord

DATASHEET AVAILABLE

Passive Components > Cable Assembly • Slug: cat6-patch-cord

The PDR U/UTP Cat 6 Patch Cord provides flexible, reliable high-speed Ethernet connectivity with Cat 6 RJ45 connectors and PVC or LSZH jacket options.

KEY FEATURES:

- U/UTP 4-pair construction with stranded 24 AWG (7×0.18 mm) conductors for flexibility and durability
- HDPE insulation for low capacitance and consistent electrical performance across all four pairs
- Cat 6 RJ45 plug termination with moulded PVC strain relief boot, colour-matched to outer jacket
- Outer jacket available in PVC or LSZH, specify at time of order to suit installation requirements

SPECIFICATIONS:

Construction	U/UTP (Unshielded Twisted Pair)
Pair Count	4 pairs
Conductor	Stranded 24 AWG (7×0.18 mm)
Insulation Material	HDPE
Insulation Diameter	0.91 ± 0.01 mm
Pair Colours	Blue/White, Orange/White, Green/White, Brown/White
Cross Filler	Yes
Connector	Cat 6 RJ45 Plug
Strain Relief Boot	Moulded PVC, colour-matched to jacket
Outer Jacket Material	PVC or LSZH (specify at order)
Jacket Thickness	0.6 ± 0.05 mm
Outer Diameter	5.8 ± 0.20 mm
Ripcord	Yes
Rated Temperature	60°C, 75°C, 90°C

#24 Bare Fiber Adapter

Passive Components > Adapters • Slug: bare-fiber-adapter

DATASHEET AVAILABLE

PDR Bare Fiber Adapters enable tool-free temporary connection of stripped bare fibers to SC, LC, FC, ST, or SMA interfaces for reliable testing, field, and lab applications.

KEY FEATURES:		SPECIFICATIONS:	
- Available in SC, LC, FC, ST, and SMA interface types to suit all common test equipment - Secure mechanical clamping mechanism holds bare fiber firmly without adhesive or tools - Compatible with both single-mode (OS1/OS2) and multimode (OM1, OM4) optical fibers - Compact, lightweight body for easy handling in lab and field environments		Adapter Types	SC, LC, FC, ST, SMA
		Compatible Fiber	Single-mode (OS1/OS2) Multimode (OM1, OM2, OM3, OM4)
		Fiber Cladding Diameter	125 µm (standard)
		Fiber Holding Mechanism	Mechanical clamp, no adhesive or tools required
		Insertion Loss	SC/FC/ST: ≤ 0.5 dB LC: ≤ 0.5 dB SMA: ≤ 1.0 dB (typical)
		Return Loss	SC/FC/ST: ≥ 20 dB LC: ≥ 20 dB SMA: ≥ 14 dB
		Mating Durability	≥ 500 mating cycles
		Housing Material	Zinc alloy body with precision ceramic / stainless insert
		Operating Temperature	20°C to +70°C
		Storage Temperature	-40°C to +85°C
		Operating Humidity	5%, 95% RH (non-condensing)
		Bare Fiber Insertion	Simple straight insertion, no stripping jig required
		Applications	Lab testing, field measurement, emergency restoration, PON/OTDR use
		Compliance	RoHS compliant

#25 Hybrid Adapter

Passive Components > Adapters • Slug: hybrid-adapter

DATASHEET AVAILABLE

Mates two dissimilar connector types (SC-FC, LC-SC, ST-FC, etc.) via a precision ceramic/phosphor-bronze sleeve, ≤0.3 dB IL and ≥50 dB RL, for SM/MM fiber in simplex or duplex.

KEY FEATURES:		SPECIFICATIONS:	
- Enables connection between different connector types, SC-FC, LC-SC, LC-FC, ST-FC, LC-ST, and more - High-precision ceramic or phosphor bronze alignment sleeve for superior optical performance - Low insertion loss ≤ 0.3 dB and high return loss ≥ 50 dB (single-mode) - Compatible with both single-mode (SM) and multi-mode (MM) fibers		Adapter Types	SC to LC, SC to FC, FC to ST, LC to ST
		Alignment Sleeve	High Precision Zirconia Ceramic (SM) / Phosphor Bronze (MM)
		Insertion Loss	≤ 0.20 dB (Typical)
		Return Loss	≥ 50 dB (UPC) / ≥ 60 dB (APC)
		Durability	> 1000 matings
		Operating Temperature	-40°C to +85°C

#26 Fiber Optic Adapter

[DATASHEET AVAILABLE](#)[Passive Components > Adapters](#) • Slug: [fiber-optic-adapter](#)

Precision passive coupler that joins two like fiber optic connectors end-to-end across a bulkhead, patch panel or enclosure, using a zirconia-ceramic or phosphor-bronze split sleeve available for all common connector standards in SM and MM.

KEY FEATURES:

- Couples two like connectors (SC, LC, FC, ST, MU, E2000, MT-RJ) for optical continuity across a panel
- High-precision zirconia-ceramic or phosphor-bronze split alignment sleeve for accurate ferrule alignment
- Low insertion loss ≤ 0.2 dB for minimal signal degradation
- High return loss ≥ 50 dB (UPC) and ≥ 60 dB (APC) for single-mode

SPECIFICATIONS:

Connector Types	SC, LC, FC, ST, MU, E2000, MT-RJ
Configuration	Simplex / Duplex / Quad
Fiber Type	Single-mode (OS1/OS2) / Multimode (OM1-OM4)
Polish	UPC / APC
Durability	≥ 1000 mating cycles

#27 Variable Fiber Attenuator

[DATASHEET AVAILABLE](#)[Passive Components > Attenuator](#) • Slug: [attenuator](#)

PDR Variable Fiber Optic Attenuator (VOA) provides continuously adjustable attenuation up to 30 dB for single-mode and multimode fiber with accurate, low-loss performance across 1260–1650 nm for telecom, CATV, data center, and test applications.

KEY FEATURES:

- Wide attenuation range of 0 to 30 dB with precise, continuous adjustment capability
- High attenuation accuracy of ± 0.5 dB for reliable signal level management
- Low insertion loss of ≤ 1.0 dB to minimise signal degradation in the optical path
- High return loss of ≥ 50 dB (SM) for excellent back-reflection suppression

SPECIFICATIONS:

Attenuation Range	0 to 30 dB (Typical)
Attenuation Accuracy	± 0.5 dB
Resolution	0.1 dB (Typical)
Insertion Loss	≤ 1.0 dB
Return Loss	≥ 50 dB (SM)
Wavelength Range	1260 to 1650 nm
Connector Type	SC / LC / FC / ST
Fiber Type	Single-mode / Multi-mode
Polarisation Dependent Loss (PDL)	≤ 0.2 dB
Repeatability	$\leq \pm 0.2$ dB
Operating Temperature	-20°C to +70°C
Storage Temperature	40°C to +85°C
Durability	≥ 1000 adjustment cycles

#28 PLC Splitter

DATASHEET AVAILABLE

Passive Components > WDM & Splitters • Slug: [plc-splitter](#)

PDR PLC Splitters offer passive optical power splitting over 1260–1650 nm in 1×N/2×N configurations with multiple packaging options for FTTx, PON, LAN, CATV, and test applications.

KEY FEATURES:

- Planar waveguide chip technology for low insertion loss, high uniformity, and excellent long-term stability
- 1×N configurations: 1×2, 1×4, 1×8, 1×16, 1×32, 1×64
- 2×N configurations: 2×2, 2×4, 2×8, 2×16, 2×32, 2×64
- Wide operating wavelength: 1260 to 1650 nm, covers all standard PON and WDM bands

SPECIFICATIONS:

Configurations	1×2, 1×4, 1×8, 1×16, 1×32, 1×64 and 2×N
Operating Wavelength	1260 nm to 1650 nm (Full Band)
Insertion Loss	≤ 3.8 dB (1×2) to ≤ 17.0 dB (1×32)
Uniformity	≤ 0.6 dB (1×2) to ≤ 1.5 dB (1×32)
Return Loss	≥ 55 dB (UPC) / ≥ 60 dB (APC)
Package Options	Steel Tube, ABS Module, LGX Cassette, 19" Rackmount

#29 CWDM Mux / DeMux Module

DATASHEET AVAILABLE

Passive Components > WDM & Splitters • Slug: [cwdm](#)

PDR CWDM Mux/DeMux offers 4/8/16-channel CWDM (1270–1610 nm) in ABS, LGX, and 1U rack formats with low loss, high isolation, and reliable optical performance.

KEY FEATURES:

- Four form factors: ABS Compact Module (two sizes), 1U Rack Equipment, and LGX Module
- Supports 4, 8, and 16 channel configurations across the CWDM grid from 1270 nm to 1610 nm
- Low insertion loss: ≤1.6 dB (4ch), ≤2.8 dB (8ch), ≤4.5 dB (16ch) for ABS Compact Module
- High adjacent channel isolation >30 dB; non-adjacent channel isolation >40 dB

SPECIFICATIONS:

CWDM Wavelength Grid	1270, 1290, 1310 ... 1570, 1590, 1610 nm (20 nm channel spacing)
Channel Configurations	4 Channel 8 Channel 16 Channel
IL, ABS Compact Module (Narrow PB)	4ch: ≤1.6 dB 8ch: ≤2.8 dB 16ch: ≤4.5 dB
IL, ABS Compact Module (Wide PB)	4ch: ≤2.0 dB 8ch: ≤3.6 dB 16ch: ≤5.0 dB
IL, 1U Rack Equipment	4ch: ≤1.8 dB 8ch: ≤3.0 dB 16ch: ≤4.8 dB
IL, LGX Module	4ch: ≤1.8 dB 8ch: ≤3.0 dB 16ch: ≤4.8 dB
Passband, ABS Compact (Narrow PB)	±0.11 nm (@-0.5 dB)
Passband, ABS Compact (Wide PB)	±6.5 nm (@-0.5 dB)
Passband, 1U Rack / LGX Module	±7.5 nm (@-0.5 dB)
Isolation, Adjacent Channel	> 30 dB
Isolation, Non-Adjacent Channel	> 40 dB
Passband Ripple	≤0.40 dB
Polarization Dependent Loss (PDL)	< 0.30 dB
Polarization Mode Dispersion (PMD)	< 0.10 ps

#30 DWDM Mux/DeMux Module

DATASHEET AVAILABLE

Passive Components > WDM & Splitters • Slug: [dwdm](#)

PDR DWDM Mux/DeMux supports 4/8/16-channel ITU-T C20–C60 DWDM (1520–1570 nm, 100 GHz) in ABS or 1U rack formats with low loss, >30 dB isolation, and >45 dB return loss.

KEY FEATURES:

- Available in compact ABS module (100×80×10 mm / 120×80×18 mm) and 1U rack equipment form factors
- Supports 4, 8, and 16 channel configurations covering ITU-T channels C20 to C60
- Low insertion loss: ≤2.0 dB (4ch), ≤3.0 dB (8ch), ≤4.8 dB (16ch) for ABS module
- Operating wavelength range 1520 to 1570 nm with 100 GHz minimum channel spacing

SPECIFICATIONS:

Channel Configurations	4 Channel 8 Channel 16 Channel
Insertion Loss, ABS Module	4ch: ≤2.0 dB 8ch: ≤3.0 dB 16ch: ≤4.8 dB
Insertion Loss, 1U Rack Equipment	4ch: ≤2.2 dB 8ch: ≤3.2 dB 16ch: ≤5.0 dB
ITU Channels	C20, C60 (C20, C21, C22 ... C58, C59, C60)
Operating Wavelength	1520 to 1570 nm
Min. Channel Spacing	100 GHz
Passband (–0.5 dB bandwidth)	±0.11 nm
Isolation, Adjacent Channel	> 30 dB
Isolation, Non-Adjacent Channel	> 40 dB
Passband Ripple	≤0.30 dB
Polarization Dependent Loss (PDL)	≤0.30 dB
Polarization Mode Dispersion (PMD)	< 0.10 ps
Return Loss	> 45 dB
Directivity	> 50 dB

#31 Fiber Optic Connector, Field Installable

DATASHEET AVAILABLE

Passive Components > Connectors • Slug: [field-connector](#)

Field-installable SC connector for tool-light on-site termination with no adhesives or polishing, fully zirconia-ceramic ferrule, ≤0.3 dB IL, ≥45 dB (PC)/≥55 dB (APC) RL, re-assemblable.

KEY FEATURES:

- Field installable without adhesives, polishing, or electrical power, ready for immediate on-site use
- Fully zirconia-ceramic ferrule: nominal 2.5 mm diameter, ±0.5 mm tolerance, ≤1 μm fiber concentricity
- Ferrule end-face geometry compliant with Telcordia GR-326 interferometric measurement standard
- Low insertion loss ≤0.3 dB and high return loss ≥45 dB (PC) / ≥55 dB (APC)

SPECIFICATIONS:

Connector Interface	SC
Polish Type	PC / APC
Body Type	Type-A / Type-B
Ferrule Material	Fully Zirconia-Ceramic
Insertion Loss	≤0.3 dB
Return Loss	≥45 dB (PC) / ≥55 dB (APC)
Compatible Cable	3.1×2 mm flat / 3 mm drop / 2.0 mm / 0.9 mm

#32

G652D Fiber Spool / Fiber Drum

DATASHEET AVAILABLE

Passive Components > Fiber Spool • Slug: fiber-spool

Precision-wound ITU-T G.652D single-mode reels with Germania-doped silica core, ultra-low attenuation under 0.200 dB/km at 1550 nm across 1310–1625 nm, individually tested and barcoded for traceability.

KEY FEATURES:

- ITU-T G.652D compliant, meets and exceeds international single-mode fiber standards
- Ultra-low attenuation under 0.200 dB/km at 1550 nm for long-haul signal integrity
- Wide operating wavelength range, optimized from 1310 nm to 1625 nm
- Dual-layer UV-curable resin coating for mechanical and environmental protection

SPECIFICATIONS:

Fiber Type	Single Mode, ITU-T G.652D
Attenuation	≤ 0.36 dB/km (1310 nm); ≤ 0.23 dB/km (1550 nm)
Cable Cutoff Wavelength	under 1260 nm
Mode Field Diameter	9.2 ±0.4 µm at 1310 nm
Cladding Diameter	125 ±0.7 µm

4.3 Cable Management Devices & Splice Enclosures (7 Products)

#33

Rack Mount Fiber Management System

DATASHEET AVAILABLE

Cable Management Devices > Fiber Termination Box (Indoor) • Slug: rack-mount-fms

PDR 19-inch Rack Mount Fiber Management System provides modular, scalable, high-density fiber management in durable 1U/2U/4U CRCA steel enclosures with sliding tray access, supporting 6–192 fiber terminations.

KEY FEATURES:

- 19-inch rack mount enclosure available in 1U, 2U, and 4U heights for scalable fiber density
- Supports 6F to 192F fiber terminations per enclosure, FC, SC, ST, and LC connector types
- Removable top cover and sliding tray mechanism for quick and easy access to service connections
- Multiple cable entry and exit openings on all models for flexible routing and installation

SPECIFICATIONS:

Rack Compatibility	19-inch standard rack mount
Available Rack Heights	1U 2U 4U
Fiber Capacity Range	6F to 192F per enclosure
Connector Types	FC / SC / ST (XX series) LC / PC (LC series)
Material	CRCA (Cold-Rolled Close-Annealed steel)
Finish	Structure finish and powder coated, 80 to 100 µm
Colour	RAL 7032 (standard) Customised on request
Cable Entry	Rear
Cable Mounting	Horizontal
Splice Tray	Hinged type ABS
Top Cover	Removable for service access
Access Mechanism	Sliding tray
Rodent Cover	Available (optional)
Dimensions (1U models)	485 mm × 315 mm × 45 mm

#34 Home Termination Box (HTB)**DATASHEET AVAILABLE**[Cable Management Devices > Fiber Termination Box \(Indoor\)](#) • Slug: **htb**

PDR Home Termination Box (HTB) is a compact wall-mounted enclosure for fiber termination, splicing, adapters, FICs, pigtails and cable storage, with bend-radius protection and splice/slack management for reliable ONT/CPE connections.

KEY FEATURES:

- Compact, low-profile design occupies minimal wall space during installation
- Inner slack storage area maintains minimum bend radius compliance at all times
- Simple, tool-free installation with included mounting screws and cable ties
- Inner tray accommodates splice sleeves for clean in-box splicing

SPECIFICATIONS:

Dimensions (L×W×H)	86 × 86 × 25 mm
Fiber Count	2 Fiber
Colour	White
Material	ABS, PP
Insertion Loss	< 0.3 dB
Return Loss	≥ 65 dB (APC) / ≥ 55 dB (UPC)
Adapter Compatibility	SC, LC (standard form factor)
Configuration	Loaded / Unloaded
Standard	IEC 61754-4

#35 Optical Fiber Wall Mount Enclosure**DATASHEET AVAILABLE**[Cable Management Devices > Fiber Termination Box \(Indoor\)](#) • Slug: **wall-mount**

PDR Optical Fiber Wall Mount Enclosure is a lockable double-door fiber termination unit for Singlemode/Multimode networks, supporting ST, FC, SC & LC adapters, loaded or unloaded options, and four cable entry points for flexible installation.

KEY FEATURES:

- Scalable port capacity, available in 12, 24, 48, and 96 port configurations
- Double-door construction with key lock for secure, access-controlled fiber termination
- Four-point cable entry (two top, two bottom) for flexible fiber routing
- Accepts Singlemode and Multimode ST, FC, SC, and LC adapter plates

SPECIFICATIONS:

Port Capacity	12, 24, 48, 96 Ports
Material & Finish	CRCA Cold Rolled Steel with 7-tank powder coating
Door Mechanism	Double Door with Master Key Cam Lock
Cable Entry	4 Knockout Grommet Ports (2 Top, 2 Bottom)
Adapter Compatibility	SC, LC Duplex, ST, FC Simplex
Mounting Type	Wall Mount / Indoor Telecom Enclosure

#36 Fiber Distribution Box (FDB)

Cable Management Devices > Fiber Termination Box (Indoor) • Slug: fdb

DATASHEET AVAILABLE

PDR-FDB24B Fiber Distribution Box is a compact, rugged FTTx termination enclosure supporting straight-joint, branch, and pass-through connectivity with up to 24 SC adaptors, two 1×8 LGX splitters, and separate cable routing for easy maintenance and fault identification.

KEY FEATURES:	SPECIFICATIONS:																												
- Accommodates 1×4, 1×8, 1×16, 2×4, 2×8, and 2×16 PLC steel tube or LGX splitter modules - IP65 protection class, suitable for both indoor and outdoor environments - Supports wall mounting and pole mounting for flexible field deployment 24 drop-cable exit ports with dedicated separation of drop cables and fiber loose tubes	<table><tr><td>Model</td><td>PDR-FDB24B</td></tr><tr><td>Dimensions (L x W x D)</td><td>320 × 275 × 105 mm</td></tr><tr><td>Weight</td><td>1.36 kg</td></tr><tr><td>Material</td><td>High Quality Engineering Plastic</td></tr><tr><td>Protection Class</td><td>IP65</td></tr><tr><td>Mounting Options</td><td>Wall Mount / Pole Mount</td></tr><tr><td>Opening Angle</td><td>> 120 degrees</td></tr><tr><td>Cable Diameters Supported</td><td>4 - 7 mm drop cable</td></tr><tr><td>Total Optical Splice Capacity</td><td>24</td></tr><tr><td>Maximum Cable Entries</td><td>2</td></tr><tr><td>Mid-Span Port</td><td>1 feeder cable entry (2 ports), 1 entrance port + 1 reserved</td></tr><tr><td>Drop Cable Exit Ports</td><td>24 ports</td></tr><tr><td>Branching & Distribution</td><td>2 in, 24 out</td></tr><tr><td>LGX Splitter Modules</td><td>2 x (1×8)</td></tr></table>	Model	PDR-FDB24B	Dimensions (L x W x D)	320 × 275 × 105 mm	Weight	1.36 kg	Material	High Quality Engineering Plastic	Protection Class	IP65	Mounting Options	Wall Mount / Pole Mount	Opening Angle	> 120 degrees	Cable Diameters Supported	4 - 7 mm drop cable	Total Optical Splice Capacity	24	Maximum Cable Entries	2	Mid-Span Port	1 feeder cable entry (2 ports), 1 entrance port + 1 reserved	Drop Cable Exit Ports	24 ports	Branching & Distribution	2 in, 24 out	LGX Splitter Modules	2 x (1×8)
Model	PDR-FDB24B																												
Dimensions (L x W x D)	320 × 275 × 105 mm																												
Weight	1.36 kg																												
Material	High Quality Engineering Plastic																												
Protection Class	IP65																												
Mounting Options	Wall Mount / Pole Mount																												
Opening Angle	> 120 degrees																												
Cable Diameters Supported	4 - 7 mm drop cable																												
Total Optical Splice Capacity	24																												
Maximum Cable Entries	2																												
Mid-Span Port	1 feeder cable entry (2 ports), 1 entrance port + 1 reserved																												
Drop Cable Exit Ports	24 ports																												
Branching & Distribution	2 in, 24 out																												
LGX Splitter Modules	2 x (1×8)																												

#37 CAT 6 Patch Panel

Cable Management Devices > Fiber Termination Box (Indoor) • Slug: cat6-panel

DATASHEET AVAILABLE

PDR Cat 6 24-Port Distribution Panel features 24 RJ-45 IDC ports with T568A/T568B support, meeting ISO/IEC 11801 and EIA/TIA standards for Gigabit and 10GBase-T structured cabling.

KEY FEATURES:

- 24-port RJ-45 panel conforming to ISO/IEC 11801 and EIA/TIA 568A & 568B standards
- Supports Cat 6 performance, suitable for Gigabit Ethernet and 10G-Base-T applications
- Metallic panel (SPCC) with fireproof ABS UL-94V0 plastic body for durability and safety
- RJ-45 jack housing: PBT / 8P8C with 50 micro-inch gold-plated contacts for reliable connectivity

SPECIFICATIONS:

Standard Compliance	ISO/IEC 11801 EIA/TIA 568A & 568B
Number of Ports	24 x RJ-45
Category	Cat 6
Panel Material	Metallic, SPCC (Steel, Cold-rolled, Coil)
Plastic Body	Fireproof ABS UL-94V0
RJ-45 Jack Housing	PBT / 8P8C, 50 micro-inch gold plated contacts
PCB	FR-4 / 1.6 mm UL-94V0
IDC Body	Fireproof ABS UL-94V0
IDC Pin Material	Phosphor Bronze, Tinned
IDC Wire Range	22 to 26 AWG solid conductor
Termination Tool	110 or Krone punch-down tool
Color Coding	T568A and T568B (Arrange: 8 7 6 3 2 1 4 5)
Insulation Resistance	> 500 MΩ (minimum)
DC Resistance	< 0.1 Ω (maximum)

#38 Heat Shrink Splice Closure

Cable Management Devices > Fiber Termination Box (Outdoor) • Slug: heat-shrink-closure

DATASHEET AVAILABLE

The PDR Fiber Optic Splice Closure is an IP68 weatherproof enclosure with heat shrink or re-enterable options, supporting up to 6 cable entries and 144 splices for outdoor fiber optic networks.

KEY FEATURES:

- IP 68 sealing protection, suitable for fully immersed and buried deployments
- Zero additional insertion loss when fibers are routed through splice trays
- Unique strength member clamp assembly prevents cable sheath movement with temperature changes
- No specialized tools required, minimizes installation time and training costs

SPECIFICATIONS:

Dimension (L × D)	530 mm × 210 mm
Weight	< 2.7 kg
Type	Butt type
Cable Sealing	Heat shrink sleeves (Heat Shrink Type) Mechanical seal (Mechanical Type)
Dome Material	PPCP

#39

Horizontal Splice Closure

Cable Management Devices > Fiber Termination Box (Outdoor) • Slug: horizontal-closure

DATASHEET AVAILABLE

Re-enterable IP65 outdoor closure for duct, aerial, and wall mounting, supporting up to 48 splices (2×24) in a UV-resistant ribbed 445×240×140 mm body.

KEY FEATURES:

- Suitable for cut, uncut, and taut sheath fiber optic cable applications
- Compatible with all standard fiber optic cable types and diameters
- Integrated sheath retention and central strength member fastening
- Ribbed closure construction using UV-resistant, lightweight plastic

SPECIFICATIONS:

Model	HTSC-TL16
Dimensions (L×W×H)	445 × 240 × 140 mm
Splice Capacity	48 splices (2 × 24)
Main Cable Ports	4 ports, max 12 mm diameter
Drop Cable Ports	16 ports, 3 mm / 3×2 drop cable
Mounting Options	Duct, Aerial, Wall
Protection Rating	IP65

4.4 Test & Measurement Equipment (7 Products)

#40

Nano OTDR

Test and Measurement Equipment > OTDR • Slug: pocket-otdr

DATASHEET AVAILABLE

The PDR Nano OTDR is a compact, smartphone-connected dual-wavelength (1310/1550 nm) OTDR for precise fiber testing, loss analysis, and real-time measurements via the Smart-OTDR app.

KEY FEATURES:

- Ultra-compact nano size, fits in a shirt pocket for true field portability
- Dual-wavelength operation at 1310 nm and 1550 nm for comprehensive fiber testing
- Wireless Bluetooth / Wi-Fi connectivity to the Smart-OTDR smartphone app (Android & iOS)
- Real-time OTDR trace display, zoom, and cursor analysis on smartphone screen

SPECIFICATIONS:

Fiber Type	Single-Mode (ITU-T G.652 / G.657)
Wavelengths	1310 nm 1550 nm
Dynamic Range	26 dB @ 1310 nm 24 dB @ 1550 nm
Distance Range	0.1 km, 60 km (auto & manual)
Distance Resolution	0.01 m
Distance Accuracy	±1 m + ±0.005% × distance
Event Dead Zone	≤ 1 m (typical, with 10 ns pulse)
Attenuation Dead Zone	≤ 4 m (typical, with 10 ns pulse)
Pulse Widths	10 ns / 30 ns / 100 ns / 300 ns / 1 μs / 3 μs
Sampling Resolution	4 cm
Loss Threshold	0.01 dB resolution
Return Loss (ORL)	Measured in dB, displayed in app
Averaging Time	5 s / 15 s / 30 s / 60 s / 180 s
Connectivity	Bluetooth 4.0 Wi-Fi 802.11 b/g/n

#41 PON Power Meter**DATASHEET AVAILABLE**Test and Measurement Equipment > Power Meter • Slug: [pon-power-meter](#)

The PDR4213B is a handheld PON Power Meter for simultaneous 1310/1490/1550 nm measurement, providing fast, accurate FTTH/GPON/EPON testing for installation and maintenance.

SPECIFICATIONS:

Wavelengths	1310 nm 1490 nm 1550 nm
Measurement Mode	Simultaneous 3-channel power measurement
Power Range	-50 dBm to +26 dBm
Resolution	0.01 dB 0.001 mW
Accuracy	+/- 0.25 dB (at reference wavelength, 23 C)
Display Units	dBm dBr mW (user selectable)
Detector Type	InGaAs photodiode
Optical Connector	SC/APC (SC/UPC adapter included)
Connector Interface	SC adapter port with 2.5 mm ferrule
Data Storage	Up to 99 measurement records (SAVE function)
Data Interface	USB 2.0 (data transfer and firmware upgrade)
Battery	3 x AA alkaline batteries
Battery Life	>= 40 hours continuous operation
Auto Power-Off	Configurable: 5 / 10 / 15 min or disabled

#42 Mini Optical Power Meter**DATASHEET AVAILABLE**Test and Measurement Equipment > Power Meter • Slug: [regular-opm](#)

Palm-sized handheld meter for FTTH continuity and link-loss testing, six calibrated wavelengths (850–1625 nm), ±0.2 dB accuracy, 0.01 dB resolution, SM/MM via 2.5 mm connector, up to 100 h battery.

KEY FEATURES:

- Palm-sized, pocket-carry handheld design for one-hand field use
- Wide measurement range with high 0.01 dB resolution and ±0.2 dB accuracy
- Six calibrated wavelengths: 850 / 1300 / 1310 / 1490 / 1550 / 1625 nm
- Supports single-mode and multimode fiber via 2.5 mm universal connector

SPECIFICATIONS:

Display Range (PDR60T)	+6 to -70 dBm (1310/1490/1550/1625 nm)
Display Range (PDR60C)	+26 to -50 dBm (1310/1490/1550/1625 nm)
Accuracy	±0.2 dB
Calibrated Wavelengths	850 / 1300 / 1310 / 1490 / 1550 / 1625 nm
Display Resolution	0.01 dB
Battery Life	Above 100 hours
Power Supply	2× AA or AC/DC adapter
Dimensions / Weight	105 × 52 × 34 mm; ~100 g

#43 Fusion Splicer PDR618H

Test and Measurement Equipment > Splicer • Slug: fusion-splicer

DATASHEET AVAILABLE

The PDR618H is a compact handheld fusion splicer for SM, MM, DS, and NZDS fibers, offering fast splicing, quick heat shrinking, and automatic ARC calibration for reliable field and lab performance.

SPECIFICATIONS:

Fiber Count	Single
Cladding Diameter	80 to 150 µm
Coating Diameter	160 to 900 µm
Typical Splice Loss	SM: 0.02 dB MM: 0.01 dB DS: 0.04 dB NZDS: 0.04 dB G.657: 0.02 dB (ITU-T Standard)
Return Loss	> 60 dB
Fiber Cleaved Length	10 to 16 mm (coating dia. < 250 µm) 16 mm (coating dia. 250 to 900 µm)
Splicing Programs	40 groups
Operating Mode	Manual / Automatic
Auto-Heating	Available
Typical Splicing Time	8 seconds
Tube Heating Time	26 seconds (60 mm and 40 mm shrinkable sleeves)
Fiber View Magnification	250× (X or Y axis) 125× (X and Y axis)
Viewing & Display	2× CMOS cameras, 4.3-inch colour LCD monitor
Splice Result Storage	4000 results

#44 **Next Gen Optical Fusion Splicer (OPM + VFL)** DATASHEET AVAILABLE

Test and Measurement Equipment > Splicer • Slug: `next-gen-splicer`

The PDR4107S is a compact core-alignment fusion splicer for SM, MM, DS, and NZDS fibers, featuring 0.02 dB splice loss, a 5-second splice time, and built-in OPM and VFL for all-in-one field splicing.

KEY FEATURES:

- Compact, lightweight design ideal for field installation and network maintenance
- Core alignment technology ensuring high-precision, low-loss fiber splicing
- Fast 5-second typical splicing time with automatic and manual operating modes
- Supports SM, MM, DS, and NZDS fiber types for maximum versatility

SPECIFICATIONS:

Applicable Fibers	SM MM DS NZDS
Fiber Count	Single
Cladding Diameter	80 to 150 µm
Coating Diameter	1 to 3 mm
Fiber Aligning Method	Automatic / Manual
Focusing Method	Four-motor automatic focusing
Typical Splice Loss	SM: 0.02 dB MM: 0.01 dB DS: 0.04 dB NZDS: 0.04 dB
Return Loss	> 60 dB
Splicing Programs	53 preset + 40 user-defined
Splicing Modes	Auto / Calibrate / Normal / Special
Operating Mode	Automatic / Manual
Typical Splicing Time	5 seconds (standard SM fiber)
Tube Heating Programs	9 preset + 24 user-defined
Tube Heating Time	15 seconds Protection sleeve: 60 mm / 40 mm

#45 **WiFi Wireless Fiber Endface Microscope** DATASHEET AVAILABLE

Test and Measurement Equipment > Endface Inspector • Slug: `wifi-wireless-fiber-endface-microscope`

Wireless/USB-C fiber endface inspector with automatic pass/fail analysis

KEY FEATURES:

- WiFi and USB Type-C data transfer with real-time video and automatic pass/fail analysis
- Compatible with Windows, Android, and iOS
- 0.42 µm resolution, 5 MP CMOS sensor, 10X optical magnification, <1 µm particle detection
- More than 50 adapter types including MPO/MTP; custom adapters available

SPECIFICATIONS:

Resolution	0.42 µm
Image Sensor	5-megapixel CMOS
Optical Magnification	10X
Field of View	512 µm × 384 µm
Signal Output	WiFi / USB Type-C
Battery Life	5 hours (rechargeable Li-ion)

#46 VFL — Visual Fault Locator

Test and Measurement Equipment > Visual Fault Locator • Slug: vfl

DATASHEET AVAILABLE

The BML209 is a compact pen-style 650 nm Visual Fault Locator (VFL) that detects fiber faults, breaks, bends, and connector defects, with 1–30 mW output options for 3 km to 10+ km testing.

KEY FEATURES:	SPECIFICATIONS:														
650 nm ±20 nm visible red laser for instant fault and break localisation - Four output power variants: 1 mW, 10 mW, 20 mW, and 30 mW for different reach requirements - Dynamic detection distance: 3 to 5 km (1 mW) up to >10 km (20/30 mW variants) - Dual operation modes: Continuous Wave (CW) and Pulsed (2 Hz) for identification	<table><tr><td>Output Power</td><td>>1 mW / >10 mW / >20 mW / >30 mW</td></tr><tr><td>Dynamic Distance</td><td>3 to 5 km / 8 to 10 km / >10 km / >10 km</td></tr><tr><td>Wavelength</td><td>650 nm ±20 nm</td></tr><tr><td>Operation Modes</td><td>Continuous Wave (CW) and Pulsed (2 Hz)</td></tr><tr><td>Connector</td><td>2.5 mm Universal adapter</td></tr><tr><td>Dimensions</td><td>180 mm (L) × 30 mm (diameter)</td></tr><tr><td>Weight</td><td>46 g</td></tr></table>	Output Power	>1 mW / >10 mW / >20 mW / >30 mW	Dynamic Distance	3 to 5 km / 8 to 10 km / >10 km / >10 km	Wavelength	650 nm ±20 nm	Operation Modes	Continuous Wave (CW) and Pulsed (2 Hz)	Connector	2.5 mm Universal adapter	Dimensions	180 mm (L) × 30 mm (diameter)	Weight	46 g
Output Power	>1 mW / >10 mW / >20 mW / >30 mW														
Dynamic Distance	3 to 5 km / 8 to 10 km / >10 km / >10 km														
Wavelength	650 nm ±20 nm														
Operation Modes	Continuous Wave (CW) and Pulsed (2 Hz)														
Connector	2.5 mm Universal adapter														
Dimensions	180 mm (L) × 30 mm (diameter)														
Weight	46 g														

4.5 Specialty Tactical Drone Optical Solutions (3 Products)

#47 UAV Fiber Optic Spool

Specialty Drones • Slug: uav-fiber-optic-spool

DATASHEET AVAILABLE

Ultra-light optical fiber payout spool for tethered UAVs and unmanned vehicles, wound with bend-insensitive G.657.A2 single-mode fiber supplied in 17 standard lengths (500 m–50 km) with an FC interface.

KEY FEATURES:	SPECIFICATIONS:						
- Ultra-light payout spool design optimized for tethered UAV and unmanned-vehicle fiber links - Bend-insensitive G657.A2 single-mode fiber for reliable high-speed payout with minimal bend loss - Optional Kevlar coating for high strength and wear resistance in harsh environments - Rated 60 N pulling force for controlled, snag-free fiber dispensing under tension	<table><tr><td>Weight</td><td>80 g/km</td></tr><tr><td>Pulling Force</td><td>60N</td></tr><tr><td>Available Lengths</td><td>500 m to 50 km (17 standard variants — see Ordering Information)</td></tr></table>	Weight	80 g/km	Pulling Force	60N	Available Lengths	500 m to 50 km (17 standard variants — see Ordering Information)
Weight	80 g/km						
Pulling Force	60N						
Available Lengths	500 m to 50 km (17 standard variants — see Ordering Information)						

#48 Sky Unit

Specialty Drones • Slug: fpv-optical-terminal

DATASHEET AVAILABLE

Aircraft-mounted transmit terminal that accepts camera composite video plus flight-controller telemetry and carries them over a single fiber at 1.25 Gbps, offered in four compact mounting sizes with a GH1.25 interface for direct integration into the drone's power and control harness.

KEY FEATURES:	SPECIFICATIONS:												
- Airborne (Type A) transmit terminal for tethered / FPV drone fiber links - Single-fiber optical transmission at 1.25 Gbps over an FC interface - Accepts composite video input and flight-controller TTL telemetry - Four mounting sizes (88×88 to 141×120 mm) for flexible airframe integration	<table><tr><td>Unit Type</td><td>Type A — Airborne Sky Unit</td></tr><tr><td>Optical Transmission Rate</td><td>1.25 Gbps</td></tr><tr><td>Optical Connector</td><td>FC (single fiber)</td></tr><tr><td>Serial / Telemetry</td><td>TTL serial (Tx/Rx) via GH1.25</td></tr><tr><td>Connector</td><td>GH1.25 (8P) back panel</td></tr><tr><td>Power Budget</td><td>5–12 V DC (via GH1.25)</td></tr></table>	Unit Type	Type A — Airborne Sky Unit	Optical Transmission Rate	1.25 Gbps	Optical Connector	FC (single fiber)	Serial / Telemetry	TTL serial (Tx/Rx) via GH1.25	Connector	GH1.25 (8P) back panel	Power Budget	5–12 V DC (via GH1.25)
Unit Type	Type A — Airborne Sky Unit												
Optical Transmission Rate	1.25 Gbps												
Optical Connector	FC (single fiber)												
Serial / Telemetry	TTL serial (Tx/Rx) via GH1.25												
Connector	GH1.25 (8P) back panel												
Power Budget	5–12 V DC (via GH1.25)												

#49 Ground unit

Specialty Drones • Slug: drone

DATASHEET AVAILABLE

Ground-side receive terminal that recovers the 1.25 Gbps single-fiber link and outputs composite video to a display while interfacing with the remote controller, image-transmission module or FPV receiver, with dial-code selectable GH1.25 or cable modes and CRSF/SBUS protocol support.

KEY FEATURES:	SPECIFICATIONS:														
- Ground-side (Type B) receive terminal for tethered / FPV drone fiber links - Single-fiber optical reception at 1.25 Gbps over an FC interface - Composite video output via BNC to display or FPV receiver - Dial-code selectable GH1.25 (W) or cable (C) interface mode	<table><tr><th>Unit Type</th><th>Type B — Ground Unit</th></tr><tr><td>Optical Transmission Rate</td><td>1.25 Gbps</td></tr><tr><td>Optical Connector</td><td>FC (single fiber)</td></tr><tr><td>Video Channel</td><td>Composite video output (BNC)</td></tr><tr><td>Serial / Telemetry</td><td>TTL serial, PPM input; CRSF / SBUS support</td></tr><tr><td>Interface Modes</td><td>GH1.25 (W) or Cable (C), dial-code selectable</td></tr><tr><td>Power Input</td><td>10–26 V DC (DC-IN); GH1.25 outputs 5 V</td></tr></table>	Unit Type	Type B — Ground Unit	Optical Transmission Rate	1.25 Gbps	Optical Connector	FC (single fiber)	Video Channel	Composite video output (BNC)	Serial / Telemetry	TTL serial, PPM input; CRSF / SBUS support	Interface Modes	GH1.25 (W) or Cable (C), dial-code selectable	Power Input	10–26 V DC (DC-IN); GH1.25 outputs 5 V
Unit Type	Type B — Ground Unit														
Optical Transmission Rate	1.25 Gbps														
Optical Connector	FC (single fiber)														
Video Channel	Composite video output (BNC)														
Serial / Telemetry	TTL serial, PPM input; CRSF / SBUS support														
Interface Modes	GH1.25 (W) or Cable (C), dial-code selectable														
Power Input	10–26 V DC (DC-IN); GH1.25 outputs 5 V														

4.6 Maintenance Tools & Cleaning Kits (3 Products)

#50 Fiber Optic Cleaner Pen

Maintenance Tools > Cleaning Tools • Slug: cleaner-pen

DATASHEET AVAILABLE

PDR Fiber Optic Cleaner Pen is a compact, single-action cleaner for female connectors, removing dust, oil, and contaminants without scratching, with anti-static microfiber technology for 800+ cleanings and optimal optical performance.

KEY FEATURES:	SPECIFICATIONS:																						
- Push-to-clean operation, easy single-motion engagement initiates the cleaning cycle 800+ cleanings per unit, cost-effective and long lasting - Anti-static resin body, prevents electrostatic charge build-up during cleaning - Densely stranded, debris-free micro-fiber cleaning element for thorough end-face cleaning	<table><tr><td>Compatible Connectors</td><td>2.5 mm: FC, SC, ST 1.25 mm: LC, MU 2.0 mm: SMPTE</td></tr><tr><td>Ferrule Tip Diameter</td><td>2.5 mm (CLEP 2.5) 1.25 mm (CLEP 1.25) 2.0 mm (CLEP 2.0)</td></tr><tr><td>Cleaning Method</td><td>Push-to-clean rotary mechanism</td></tr><tr><td>Cleaning Rotation</td><td>180° full sweep per actuation</td></tr><tr><td>Cleanings Per Unit</td><td>800+ cleanings</td></tr><tr><td>Cleaning Element</td><td>Densely stranded micro-fiber, debris-free</td></tr><tr><td>Tip Design</td><td>Extendable, reaches recessed connectors</td></tr><tr><td>Engagement Indicator</td><td>Audible click on correct engagement</td></tr><tr><td>Body Material</td><td>Anti-static resin</td></tr><tr><td>Operation</td><td>Single push motion, no batteries required</td></tr><tr><td>Disposal</td><td>Disposable, replace when cleanings are exhausted</td></tr></table>	Compatible Connectors	2.5 mm: FC, SC, ST 1.25 mm: LC, MU 2.0 mm: SMPTE	Ferrule Tip Diameter	2.5 mm (CLEP 2.5) 1.25 mm (CLEP 1.25) 2.0 mm (CLEP 2.0)	Cleaning Method	Push-to-clean rotary mechanism	Cleaning Rotation	180° full sweep per actuation	Cleanings Per Unit	800+ cleanings	Cleaning Element	Densely stranded micro-fiber, debris-free	Tip Design	Extendable, reaches recessed connectors	Engagement Indicator	Audible click on correct engagement	Body Material	Anti-static resin	Operation	Single push motion, no batteries required	Disposal	Disposable, replace when cleanings are exhausted
Compatible Connectors	2.5 mm: FC, SC, ST 1.25 mm: LC, MU 2.0 mm: SMPTE																						
Ferrule Tip Diameter	2.5 mm (CLEP 2.5) 1.25 mm (CLEP 1.25) 2.0 mm (CLEP 2.0)																						
Cleaning Method	Push-to-clean rotary mechanism																						
Cleaning Rotation	180° full sweep per actuation																						
Cleanings Per Unit	800+ cleanings																						
Cleaning Element	Densely stranded micro-fiber, debris-free																						
Tip Design	Extendable, reaches recessed connectors																						
Engagement Indicator	Audible click on correct engagement																						
Body Material	Anti-static resin																						
Operation	Single push motion, no batteries required																						
Disposal	Disposable, replace when cleanings are exhausted																						

#51 Fiber Optic Cleaner Pen MPO

Maintenance Tools > Cleaning Tools • Slug: mpo-cleaner

DATASHEET AVAILABLE

PDR MPO/MTP Fiber Optic Cleaner Pen provides fast, dry cleaning of male and female MPO/MTP connectors, cleaning all 12 fibers in one push, fitting adapters and exposed ferrules, and delivering up to 500 cleaning cycles.

KEY FEATURES:

- Pen-style handheld design with narrow profile for tightly-spaced MPO adapter panels
- Cleans all 12 fiber end-faces simultaneously in a single push-and-release action
- Dry mechanical cleaning, no alcohol or wet chemicals required
- Compatible with male (with guide pins) and female (without guide pins) MPO/MTP ferrules

SPECIFICATIONS:

Compatible Connectors	MPO / MTP
Connector Standard	FOCIS-5 (IEC 61754-7)
Fiber Mode	Single-mode (SM) and Multimode (MM)
Ferrule Configuration	Male (with guide pins) and Female (without guide pins)
Polish Type	PC (Physical Contact) and APC (Angled Physical Contact)
Cleaning Method	Dry mechanical, no alcohol or solvents required
Fibers Cleaned per Stroke	12 fibers simultaneously
Cleaning Capacity	Up to 500 cleanings per unit
Operation	One-handed push-and-release mechanism
Cleaning Compatibility	Adapter-mounted connectors and exposed jumper ferrules
Contaminant Types	Dust, oils, fingerprints, and particulate matter
Cleaning Standard	IEC 61300-3-35 (fiber end-face cleanliness)
Design	Pen-style, handheld, narrow body for panel access

#52Cassette Cleaner

Maintenance Tools > Cleaning Tools • Slug: cassette-cleaner

DATASHEET AVAILABLE

PDR Fiber Optic Connector Cleaner: Compact handheld cleaner for SC, FC, ST, MU, LC, MT, D4, DIN & E2000 connectors, using non-alcohol superfine fiber tape for scratch-free, residue-free cleaning with 500+ cycles per replaceable cartridge.

KEY FEATURES:	SPECIFICATIONS:																		
- Compatible with SC, FC, ST, MU, LC, MT, D4, DIN, and E2000 fiber optic connectors - Special non-alcohol superfine fiber tape for safe, residue-free end-face cleaning - Anti-static design prevents re-contamination after each cleaning stroke - Over 500 cleaning cycles per tape cartridge, low cost per clean	<table><tr><td>Compatible Connectors</td><td>SC, FC, ST, MU, LC, MT, D4, DIN, E2000</td></tr><tr><td>Cleaning Medium</td><td>Non-alcohol superfine fiber tape</td></tr><tr><td>Cleaning Times per Cartridge</td><td>500+ cycles</td></tr><tr><td>Replacement Cartridge</td><td>Available (field-replaceable)</td></tr><tr><td>Anti-static</td><td>Yes</td></tr><tr><td>Cleaning Action</td><td>One-push mechanical advance</td></tr><tr><td>Dimensions (L×W×H)</td><td>115 mm × 79 mm × 32 mm</td></tr><tr><td>Weight</td><td>Lightweight, suitable for field carry</td></tr><tr><td>Solvents / Liquids Required</td><td>None</td></tr></table>	Compatible Connectors	SC, FC, ST, MU, LC, MT, D4, DIN, E2000	Cleaning Medium	Non-alcohol superfine fiber tape	Cleaning Times per Cartridge	500+ cycles	Replacement Cartridge	Available (field-replaceable)	Anti-static	Yes	Cleaning Action	One-push mechanical advance	Dimensions (L×W×H)	115 mm × 79 mm × 32 mm	Weight	Lightweight, suitable for field carry	Solvents / Liquids Required	None
Compatible Connectors	SC, FC, ST, MU, LC, MT, D4, DIN, E2000																		
Cleaning Medium	Non-alcohol superfine fiber tape																		
Cleaning Times per Cartridge	500+ cycles																		
Replacement Cartridge	Available (field-replaceable)																		
Anti-static	Yes																		
Cleaning Action	One-push mechanical advance																		
Dimensions (L×W×H)	115 mm × 79 mm × 32 mm																		
Weight	Lightweight, suitable for field carry																		
Solvents / Liquids Required	None																		

5. PDR 5-Segment Ordering Code Standard

PDR assemblies are specified using a deterministic 5-segment part number:

[Assembly Code] – [End #1 Connector] – [End #2 Connector] – [Cable Construction] – [Length Code]

Example: S1-L6-C2-01-0030

└ S1: Single Mode G.652D Fiber Core (9/125 μm)

└ L6: End #1 = LC/APC (Angled Polish, Green Housing)

└ C2: End #2 = SC/UPC (Flat Polish, Blue Housing)

└ 01: Cable Construction = 3.0mm Simplex LSZH Jacket

└ 0030: Length Code = 3.0 Meters (30 × 0.1m)

Segment 1: Assembly (Fiber Core)

CODE	FIBER CORE SPECIFICATION
S1	Single Mode G.652D Standard (9/125 μm)
S2	Single Mode G.657.A1 Bend-Insensitive
S3	Single Mode G.657.A2/B3 Ultra-Bend
M1	Multimode OM1 (62.5/125 μm)
M2	Multimode OM2 (50/125 μm)
M3	Multimode OM3 Laser-Optimized (50/125 μm)
M4	Multimode OM4 High-Speed VCSEL
M5	Multimode OM5 Wideband SWDM

Segment 2 & 3: Connector Codes

CODE	CONNECTOR & POLISH TYPE
00	Pigtail (Bare Cleaved End, No Connector)
L2 / L6	LC/UPC (Blue) / LC/APC (Green)
C2 / C6	SC/UPC (Blue) / SC/APC (Green)
F2 / F6	FC/UPC / FC/APC (Threaded)
T2	ST/UPC Bayonet
L8 / C8	LC / SC Multimode (Beige/Aqua)
M5 / M6	MPO 12F Male (Pins) / Female (Pinless)
E2 / E6	E2000 UPC / APC Spring Shutter

Segment 4: Cable Construction

CODE	CONSTRUCTION & DIAMETER
01 / 07	Simplex 3.0mm LSZH (SM / MM)
02 / 08	Duplex Zipcord 3.0mm (SM / MM)
03 / 20	900µm Tight Buffer (SM / MM)
09 / 21	Simplex 2.0mm LSZH (SM / MM)
16 / 22	Duplex Zipcord 2.0mm (SM / MM)
31 / 32	Armored Duplex (8.0mm SM / 7.0mm MM)
43	FTTH Bow-Type Drop Cable 3.0mm
51 / 54	12-Fiber Fanout Assembly (SM / MM)

Segment 5: Length Code Formula

Formula: `Code = round(Length in Meters × 10)` (4 digits)

LENGTH	4-DIGIT CODE & APPLICATION
0.5 Meter	0005 • Ultra-short rack patch
1.0 Meter	0010 • Standard intra-rack interconnect
2.0 Meters	0020 • Server to ToR switch patch
3.0 Meters	0030 • Standard patch cord length
5.0 Meters	0050 • Inter-rack structured patch
10.0 Meters	0100 • Row-to-row backbone link
50.0 Meters	0500 • Campus / riser backbone
100.0 Meters	1000 • Long-span distribution run

6. Interactive Web Tools & Digital RFQ Pipeline

3D Cable Configurator (`/cable-configurator`)

Procedurally generates 3D WebGL meshes using **Three.js** with real-time mouse rotation, dynamic lighting, color-coded standard jackets, and direct Quote Cart sync.

Optical Link Budget Calculator

`Total Loss = (Dist [km] × Fiber Loss) + (Conn Count × 0.35 dB)`

• ≤ 10dB: Excellent • 10–15dB: Good • 15–20dB: Fair • >20dB: Poor

Multi-Tier RFQ Submission Engine

- **Cart Aggregation:** Guest SHA-256 session hash stored in `LocalStorage`.
- **Database Insertion:** Atomic PostgreSQL RPC (``submit_quote_request``).
- **Google Sheets Sync:** Real-time row append via Google Sheets API v4.
- **Offline Buffer:** Auto-saves to `LocalStorage` if network drops.

Admin Panel Operations (`/admin`)

- **Single Direct Login:** Authenticated via Supabase Auth + 30-min idle timeout.
- **Product CRUD:** In-place edit, delete, and add with live image/PDF upload.
- **Instant Cache Bust:** Website updates in real time for all visitors.
- **Lead Management:** Live RFQ status flow (``New`` → ``In Progress`` → ``Quoted``).

7. Enterprise SEO, Structured Data & Prerendering Engine

Schema.org JSON-LD (8 Implementations)

- **Organization & LocalBusiness:** Legal entity, geo-coordinates, GSTIN, and verified social links.
- **Product (52 SKUs):** Injects name, brand, image, and dynamic `additionalProperty` spec tables.
- **Breadcrumbs & Sitelinks:** `BreadcrumbList` hierarchy + `WebSite` SearchAction.

Prerendering, Sitemap & Meta Directives

- **Static Prerendering:** Static HTML snapshots for WhatsApp, LinkedIn & search bots.
- **Geo-Targeting:** `geo.region: IN-MH` , `geo.position: 19.0144;72.8295` .
- **hreflang & Sitemap:** `en-in` , `en` , `x-default` + dynamic 69+ URL sitemap.
- **Noindex Safety:** Internal `/admin` routes inject `noindex, nofollow` .

8. Complete 52-Product Quick Reference Master Matrix

#	PRODUCT NAME	CATEGORY	CORE TECHNICAL SPECIFICATIONS
1	Active Optical Cable (AOC)	Active Optical Cable (AOC)	Data Rate: 10G / 25G / 40G / 100G / 400G; Form Factor: SFP+, SFP28, QSFP+, QSFP28, QSFP-DD; Cable Type: Multimode Fiber (OM3 / OM4)
2	Variable Fiber Attenuator	Attenuator	Attenuation Range: 0 to 30 dB (Typical); Attenuation Accuracy: ± 0.5 dB; Resolution: 0.1 dB (Typical)
3	Bare Fiber Adapter	Adapters	Adapter Types: SC, LC, FC, ST, SMA; Compatible Fiber: Single-mode (OS1/OS2) Multimode (OM1, OM2, OM3, OM4); Fiber Cladding Diameter: 125 µm (standard)
4	Cassette Cleaner	Cleaning Tools	Compatible Connectors: SC, FC, ST, MU, LC, MT, D4, DIN, E2000; Cleaning Medium: Non-alcohol superfine fiber tape; Cleaning Times per Cartridge: 500+ cycles
5	CAT 6 Patch Panel	Fiber Termination Box (Indoor)	Standard Compliance: ISO/IEC 11801 EIA/TIA 568A & 568B; Number of Ports: 24 x RJ-45; Category: Cat 6
6	CAT 6 Patch Cord	Cable Assembly	Construction: U/UTP (Unshielded Twisted Pair); Pair Count: 4 pairs; Conductor: Stranded 24 AWG (7×0.18 mm)
7	Fiber Optic Cleaner Pen	Cleaning Tools	Compatible Connectors: 2.5 mm: FC, SC, ST 1.25 mm: LC, MU 2.0 mm: SMPTE; Ferrule Tip Diameter: 2.5 mm (CLEP 2.5) 1.25 mm (CLEP 1.25) 2.0 mm (CLEP 2.0); Cleaning Method: Push-to-clean rotary mechanism
8	CPRI Patchcord	Cable Assembly	Cable Diameter: 5.0 mm 7.0 mm 8.0 mm (±0.3 mm); Connector Housing Types: ODVA IP FX PDLC NSN; Ferrule / Connector Types: MPO/MTP LC SC
9	CWDM Mux / DeMux Module	WDM & Splitters	CWDM Wavelength Grid: 1270, 1290, 1310 ... 1570, 1590, 1610 nm (20 nm channel spacing); Channel Configurations: 4 Channel 8 Channel 16 Channel; IL, ABS Compact Module (Narrow PB): 4ch: ≤1.6 dB 8ch: ≤2.8 dB 16ch: ≤4.5 dB
10	Direct Attached Cable (DAC)	Direct Attached Cable (DAC)	Cable Type: Passive twinax copper; Connector Type: SFP+ (both ends); Nominal Data Rate: 10.3125 Gbps
11	Ground unit	Specialty Drones	Unit Type: Type B — Ground Unit; Optical Transmission Rate: 1.25 Gbps; Optical Connector: FC (single fiber)
12	DWDM Mux/DeMux Module	WDM & Splitters	Channel Configurations: 4 Channel 8 Channel 16 Channel; Insertion Loss, ABS Module: 4ch: ≤2.0 dB 8ch: ≤3.0 dB 16ch: ≤4.8 dB; Insertion Loss, 1U Rack Equipment: 4ch: ≤2.2 dB 8ch: ≤3.2 dB 16ch: ≤5.0 dB
13	FanOut Patch Cords	Cable Assembly	MPO Insertion Loss, Typical: SM Standard: 0.35 dB SM Low Loss: 0.20 dB MM Standard: 0.35 dB MM Low Loss: 0.20 dB; MPO Insertion Loss, Maximum: SM Standard: 0.75 dB SM Low Loss: 0.35 dB MM Standard: 0.60 dB MM Low Loss: 0.35 dB; LC / SC / ST / FC Insertion Loss, Typical: SM Standard: 0.3 dB SM Low Loss: 0.2 dB MM Standard: 0.3 dB MM Low Loss: 0.2 dB
14	Fiber Distribution Box (FDB)	Fiber Termination Box (Indoor)	Model: PDR-FDB24B; Dimensions (L x W x D): 320 × 275 × 105 mm; Weight: 1.36 kg
15	G652D Fiber Spool / Fiber Drum	Fiber Spool	Fiber Type: Single Mode, ITU-T G.652D; Attenuation: ≤ 0.36 dB/km (1310 nm); ≤ 0.23 dB/km (1550 nm); Cable Cutoff Wavelength: under 1260 nm

#	PRODUCT NAME	CATEGORY	CORE TECHNICAL SPECIFICATIONS
16	Fiber Optic Connector, Field Installable	Connectors	Connector Interface: SC; Polish Type: PC / APC; Body Type: Type-A / Type-B
17	Fiber Optic Patch Cords & Pigtails	Cable Assembly	Cable Diameter: 0.9 mm 2.0 mm 3.0 mm (or custom); Cable Configuration: Simplex Duplex Fanout (up to 24F); SM Fiber Type: OS2 G.657A1 G.657A2
18	Fusion Splicer PDR618H	Splicer	Fiber Count: Single; Cladding Diameter: 80 to 150 µm; Coating Diameter: 160 to 900 µm
19	Heat Shrink Splice Closure	Fiber Termination Box (Outdoor)	Dimension (L × D): 530 mm × 210 mm; Weight: < 2.7 kg; Type: Butt type
20	Horizontal Splice Closure	Fiber Termination Box (Outdoor)	Model: HTSC-TL16; Dimensions (L×W×H): 445 × 240 × 140 mm; Splice Capacity: 48 splices (2 × 24)
21	Hybrid Adapter	Adapters	Adapter Types: SC to LC, SC to FC, FC to ST, LC to ST; Alignment Sleeve: High Precision Zirconia Ceramic (SM) / Phosphor Bronze (MM); Insertion Loss: ≤ 0.20 dB (Typical)
22	Mode Conditioning Patchcord	Cable Assembly	Cable Plant: 62.5/125 or 50/125 Multimode Fiber (MMF); Insertion Loss (MMF–SMF): ≤ 2.1 dB (50/125); ≤ 2.6 dB (62.5/125); Return Loss: ≥ 50 dB (UPC); ≥ 65 dB (APC)
23	MPO Cable Assembly	Cable Assembly	Fiber Mode: SM: OS2 MM: OM3, OM4, OM5; Fiber Count: 4F 8F 12F 24F 48F 72F 96F 144F Custom; Cable Type: Jacketed Ribbon Bare Ribbon Round
24	Fiber Optic Cleaner Pen MPO	Cleaning Tools	Compatible Connectors: MPO / MTP; Connector Standard: FOCIS-5 (IEC 61754-7); Fiber Mode: Single-mode (SM) and Multimode (MM)
25	Next Gen Optical Fusion Splicer (OPM + VFL)	Splicer	Applicable Fibers: SM MM DS NZDS; Fiber Count: Single; Cladding Diameter: 80 to 150 µm
26	Optical Line Protection System (OLPS)	Optical Line Protection System (OLPS)	Protection Architecture: 1+1 1:1 1-1; Form Factor: 1U rack-mountable (19-inch); Working Wavelength: 1310 nm or 1550 nm (selectable)
27	PLC Splitter	WDM & Splitters	Configurations: 1×2, 1×4, 1×8, 1×16, 1×32, 1×64 and 2×N; Operating Wavelength: 1260 nm to 1650 nm (Full Band); Insertion Loss: ≤ 3.8 dB (1×2) to ≤ 17.0 dB (1×32)
28	Nano OTDR	OTDR	Fiber Type: Single-Mode (ITU-T G.652 / G.657); Wavelengths: 1310 nm 1550 nm; Dynamic Range: 26 dB @ 1310 nm 24 dB @ 1550 nm
29	POF — Plastic Optical Fiber Patchcord	Cable Assembly	Fiber Standard: IEC 60793-2-40 Class A4a (SI-POF); Numerical Aperture: 0.5; Attenuation at 650 nm: ≤ 0.19 dB/m
30	PON Power Meter	Power Meter	Wavelengths: 1310 nm 1490 nm 1550 nm; Measurement Mode: Simultaneous 3-channel power measurement; Power Range: -50 dBm to +26 dBm
31	Rack Mount Fiber Management System	Fiber Termination Box (Indoor)	Rack Compatibility: 19-inch standard rack mount; Available Rack Heights: 1U 2U 4U; Fiber Capacity Range: 6F to 192F per enclosure
32	FTTX Smart Bullet Drop Cable	Cable Assembly	Connector Type: SC/APC bullet-shaped (near-end) / SC/APC (far-end); Bullet Connector Diameter: Ø5.9 mm; Operating Wavelength: 1260 to 1650 nm
33	Mini Optical Power Meter	Power Meter	Display Range (PDR60T): +6 to -70 dBm (1310/1490/1550/1625 nm); Display Range (PDR60C): +26 to -50 dBm (1310/1490/1550/1625 nm); Accuracy: ±0.2 dB
34	100G & 100G BIDI	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
35	SFP+ 10G BIDI	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
36	SFP+ 10G Dual Fiber	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
37	1G SFP BIDI	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
38	25G & 25G BIDI	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
39	400G	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
40	40G	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km

#	PRODUCT NAME	CATEGORY	CORE TECHNICAL SPECIFICATIONS
41	Copper SFP+	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
42	Smart SFP Transceiver	SFP Transceivers	Interface: 1G / 10G SFP; Protocol: IEEE 802.3ah / 802.1ag; Diagnostics: DDM/DOM
43	SMPTE Cable Assembly	Cable Assembly	Fiber Type: Single-mode (9/125 µm); Multimode (50/125, 62.5/125, OM3); Connector Options: SC, LC, FC, ST, MU, MTRJ, SMA, E2000; Insertion Loss: ≤ 0.30 dB typical; ≤ 0.50 dB max
44	VFL — Visual Fault Locator	Visual Fault Locator	Output Power: >1 mW / >10 mW / >20 mW / >30 mW; Dynamic Distance: 3 to 5 km / 8 to 10 km / >10 km / >10 km; Wavelength: 650 nm ±20 nm
45	Optical Fiber Wall Mount Enclosure	Fiber Termination Box (Indoor)	Port Capacity: 12, 24, 48, 96 Ports; Material & Finish: CRCA Cold Rolled Steel with 7-tank powder coating; Door Mechanism: Double Door with Master Key Cam Lock
46	Bypass Switch	Bypass Switch	Working Wavelength: 1310 nm / 1550 nm (selectable); Fiber Type: SM 9/125 µm (ITU-T G.652); Channels: Up to 4 independent channels
47	Home Termination Box (HTB)	Fiber Termination Box (Indoor)	Dimensions (L×W×H): 86 × 86 × 25 mm; Fiber Count: 2 Fiber; Colour: White
48	1G SFP Dual Fiber	SFP Transceivers	SFP: Up to 2.67 Gbps · SM/MM · up to 80 km; SFP+: Up to 11.3 Gbps · SM/MM · up to 40 km; SFP28: 25.78 Gbps · SM/MM · up to 10 km
49	UAV Fiber Optic Spool	Specialty Drones	Weight: 80 g/km; Pulling Force: 60N; Available Lengths: 500 m to 50 km (17 standard variants — see Ordering Information)
50	Sky Unit	Specialty Drones	Unit Type: Type A — Airborne Sky Unit; Optical Transmission Rate: 1.25 Gbps; Optical Connector: FC (single fiber)
51	Fiber Optic Adapter	Adapters	Connector Types: SC, LC, FC, ST, MU, E2000, MT-RJ; Configuration: Simplex / Duplex / Quad; Fiber Type: Single-mode (OS1/OS2) / Multimode (OM1-OM4)
52	WiFi Wireless Fiber Endface Microscope	Endface Inspector	Resolution: 0.42 µm; Image Sensor: 5-megapixel CMOS; Optical Magnification: 10X

9. Corporate Sign-off, Warranty & Mission Statement

PDR Videotronics India Pvt. Ltd. certifies that all manufactured optical fiber products, active interfaces, cable assemblies, and enclosures conform to international telecommunication standards (Telcordia, IEC, TIA/EIA, ITU-T, and RoHS).

12-Month Comprehensive Manufacturer Warranty

All passive cable assemblies, patch panels, splice closures, and termination boxes carry a standard 12-month warranty against material and manufacturing defects.

Issued By: Quality Assurance & Engineering Directorate
PDR Videotronics India Pvt. Ltd. • Mumbai, Maharashtra, India

Authorized Signatory: PDR Technical Operations
www.pdrworld.com • info@pdrworld.com • +91-22-24306494